Simulating Andes Hantavirus Transmission in a Structured Coastal Community

Simulation and Modeling, Bocconi University, Milan, Italy

Authors:
Agazzi Alice 3194647, Dell'Amico Greta 3197417, Lazzeri Marco 3184584, Tomasoni Michela 320091

Running title: ANDV Transmission and Control on Contact Networks
Word count (main text): 4163

Target journal: Epidemics

Abstract

Andes virus (ANDV) is the only hantavirus for which person-to-person transmission has been documented, primarily through close household, social, and healthcare contacts. We investigated a hypothetical introduction of ANDV into an Italian coastal community following cruise-linked importation. The aim was not to predict a real outbreak, but to evaluate how contact structure, pre-symptomatic transmission, and reactive interventions influence epidemic risk.

A deterministic SEIR/SEIQR framework and a stochastic network model were developed using a close-contact network combining household clusters and age-structured extra-household contacts. Transmission parameters were calibrated to the 2018–2019 Epuyén outbreak in Argentina. The baseline model assumed an 18-day incubation period, a 5-day infectious period, and conservative pre-symptomatic infectiousness equal to that of symptomatic cases ($\varepsilon = 1$).

Without intervention, the deterministic model predicted a large outbreak with an high attack rate. Rapid isolation of symptomatic cases reduced epidemic burden but remained insufficient for reliable containment because transmission occurring before symptom onset sustained epidemic growth. Analytical results identified a critical contact-tracing efficiency threshold of approximately $\eta^* = 0.75$ for epidemic control. Network simulations highlighted the importance of stochastic extinction: only 39% of introductions generated major outbreaks, but established epidemics remained substantial. Symptomatic isolation reduced the probability of a major outbreak, while contact tracing and quarantine reduced it to further and lowered average attack rates by approximately two-thirds.

The results suggest that outbreak preparedness for ANDV should prioritise rapid contact tracing, high compliance with quarantine measures, and the early identification of exposed contacts. Because transmission may occur before symptom onset, interventions targeting exposed individuals are likely to provide greater epidemiological benefit than reliance on symptomatic isolation alone.

Keywords: Andes virus; Hantavirus cardiopulmonary syndrome; Contact tracing; Network epidemiology; Stochastic epidemic modelling; Pre-symptomatic transmission

1. Introduction

Hantavirus cardiopulmonary syndrome is a rare but severe zoonotic disease. Most hantaviruses are acquired through exposure to infected rodent excreta and do not sustain human-to-human transmission. Andes virus is the major exception. Epidemiological and genomic evidence from Argentina and Chile shows that ANDV can propagate through close human contacts, particularly within households, between sexual partners, in healthcare exposure, and after crowded social events [1–3]. This feature makes ANDV a suitable case for a network-based intervention model, because the relevant transmission process is shaped by clustered contacts rather than by homogeneous population mixing [4]. Because detailed empirical contact networks are unavailable, a synthetic close-contact network was constructed to reproduce key structural features of transmission, including household clustering and stable extra-household interactions.

The motivating scenario considers a hypothetical introduction of Andes virus (ANDV) into an Italian coastal community following the disembarkation of infected cruise passengers. While inspired by a documented cruise-associated ANDV cluster [5], the outbreak simulated in this study is entirely fictional. The framework should therefore be interpreted as a stress test designed to explore how a pathogen with the known epidemiological characteristics of ANDV might spread within a stylised Italian population and to assess which intervention strategies are most effective at preventing sustained transmission.

The analysis is guided by three research questions. First, how does an ANDV outbreak evolve in a population characterised by strong household clustering when no control measures are implemented? Second, to what extent can the rapid isolation of symptomatic individuals reduce the epidemic peak and overall attack rate when diagnosis is subject to realistic delays? Third, is there a critical combination of contact-tracing efficiency and population compliance above which transmission becomes subcritical in the mean-field model, and how does this threshold translate to outbreak risk on an explicit contact network?

Because outbreaks are initiated from a single imported case, stochastic extinction is expected to play an important role. The network simulations therefore focus not only on outbreak size, but also on the probability that transmission becomes established. Consequently, both epidemic burden and the probability of a major outbreak are used as outcome measures throughout the study.

The strongest empirical anchor is the 2018-2019 Epuyen outbreak in Argentina [2]. After a single zoonotic introduction, transmission was reconstructed through clinical investigation, contact tracing, serology, and sequencing. The outbreak involved 34 confirmed infections and 11 deaths, and transmission was disproportionately driven by three symptomatic individuals who attended crowded social events. Before public-health control measures, the median reproductive number was estimated at 2.12; after isolation and quarantine were enforced, it declined to 0.96 [2]. This outbreak therefore provides a well-documented empirical basis for calibrating the baseline reproduction number and for defining a reference benchmark against which the effectiveness of control measures can be evaluated.

The temporal natural history of ANDV also differs from common respiratory infections. The incubation period is longer, with a median close to 18 days and a range extending to approximately six weeks [6]. Evidence from household-contact studies in Chile found that ANDV RNA could be detected in blood samples 5–15 days before symptoms appeared or before antibodies became detectable [7]. In another case, the virus was isolated from a child before the onset of illness, although the child later developed symptoms [8]. While these findings do not demonstrate that pre-symptomatic transmission is always effective, they suggest that infected individuals may be capable of transmitting

the virus before becoming symptomatic. To account for this possibility, the model assumes pre-symptomatic infectiousness during the exposed stage, represented by epsilon; the baseline model uses epsilon = 1.0 as a conservative assumption.

Contact structure is a central modelling issue. Household-contact studies in Chile found the highest risk among sex partners and close household contacts [7], and a Chilean cluster provided evidence of household and nosocomial transmission [3]. More general contact-survey work shows that physical, home, long-duration, and frequent contacts explain close-contact infection patterns better than broad social contacts [9]. The network therefore combines household cliques with stable extra-household close contacts; the Prem Italy matrix is used only to weight extra-household close-contact sampling by age, not to create a second transmission network.

The project also draws on contact-tracing models developed for COVID-19 and other directly transmitted infections [10,11]. Although these models concern different pathogens, they illustrate a common principle: tracing is most effective when exposed contacts are identified and isolated before they can transmit. By contrast, isolating only symptomatic cases is less effective when transmission can occur before symptom onset. This insight motivates both the analytical η–ψ threshold analysis and the subsequent network-based evaluation of detection-triggered contact tracing.

2. Methods

2.1 Population and contact network

A synthetic population of 10,000 individuals was generated to represent a stylised Italian coastal community. Individuals were assigned to five broad age groups using demographic proportions derived from Italian population statistics. Household sizes were sampled from the 2021 Italian census distribution [12], and all members of the same household were connected through complete household cliques.

The contact network was designed to represent the close and repeated interactions considered most relevant for person-to-person transmission of Andes virus (ANDV). Two layers of contacts were included. The first layer consisted of household connections, reflecting exposure among co-residing individuals. The second layer consisted of stable extra-household close contacts representing interactions such as friendships, relatives, or regular workplace contacts. The mean number of extra-household close contacts was calibrated using physical and prolonged-contact information derived from the POLYMOD study [9]. Age-specific mixing patterns were incorporated through the synthetic Italy contact matrix developed by Prem et al. [13], which provides empirically informed estimates of social contact frequencies derived from demographic structure, household composition, school attendance, workplace participation, and community interactions. The original matrix was aggregated into five age categories (0–4, 5–17, 18–39, 40–64, and 65+ years) to match the demographic structure of the network model while preserving the dominant age-assortative mixing patterns.

The matrix was used only to weight the selection of extra-household contacts across age groups and therefore influenced who was likely to interact with whom. It was not used as a transmission matrix. Instead, disease transmission was governed independently by the epidemiological parameters and calibrated to the target reproduction number derived from the Epuyén outbreak [2]. Consequently, the contact matrix defines patterns of social interaction, whereas transmission dynamics are determined by the disease model itself.

Because detailed empirical contact networks are unavailable for the hypothetical study population, the resulting graph should be interpreted as a synthetic approximation designed to reproduce key structural features of close-contact transmission [14,15], including household clustering, repeated interactions, and age-assortative mixing.

2.2 Transmission and disease progression

Transmission dynamics were represented using compartmental epidemic models and corresponding stochastic network simulations. The baseline formulation consisted of a susceptible–exposed–infectious–removed (SEIR) structure in which individuals progressed through four epidemiological states: susceptible (S), exposed (E), infectious (I), and removed (R).

Model parameters were selected from published evidence on Andes virus outbreaks. The baseline population consisted of 10,000 individuals with a pre-intervention reproduction number of $R_0 = 2.12$, consistent with estimates from the 2018 Epuyén outbreak [2]. The median incubation period was assumed to be 18 days, corresponding to an exposed-to-infectious transition rate $\sigma = 1/18$ day^{-1}. The symptomatic infectious period was assumed to last 5 days on average, corresponding to a recovery rate $\gamma = 1/5$ day^{-1}. The relative infectiousness of exposed individuals was set to $\varepsilon = 1.0$, a conservative assumption that treats pre-symptomatic and symptomatic individuals as equally infectious. The transmission parameter β was calibrated analytically to reproduce the target reproduction number under these assumptions.

The baseline deterministic dynamics are described by:

$$\frac{dS}{dt} = -\beta \frac{(\varepsilon E + I)}{N} S$$
$$\frac{dE}{dt} = \beta \frac{(\varepsilon E + I)}{N} S - \sigma E$$
$$\frac{dI}{dt} = \sigma E - \gamma I$$
$$\frac{dR}{dt} = \gamma I$$

where β is the transmission rate, σ is the rate at which exposed individuals become infectious, and γ is the recovery rate. The parameter ε scales the infectiousness of exposed individuals relative to symptomatic infectious individuals.

Intervention analyses extended the baseline model to include isolation and quarantine compartments. Infectious individuals could be removed from transmission following diagnosis and isolation, while traced exposed contacts could enter quarantine before becoming infectious. The resulting susceptible–exposed–infectious–quarantined–removed (SEIQR) formulation is:

$$\frac{dS}{dt} = -\beta \frac{(\varepsilon E + I)}{N} S$$
$$\frac{dE}{dt} = \beta \frac{(\varepsilon E + I)}{N} S - \sigma E - \theta E$$
$$\frac{dE_Q}{dt} = \theta E - \sigma E_Q$$
$$\frac{dI}{dt} = \sigma E - (\gamma + \delta) I$$
$$\frac{dQ}{dt} = \delta I + \sigma E_Q - \gamma Q$$
$$\frac{dR}{dt} = \gamma (I + Q)$$

In this framework, δ denotes the isolation rate of infectious individuals and corresponds to an average diagnostic delay $\tau = 1/\delta$. The parameter θ represents the rate at which exposed contacts are successfully traced and quarantined. The compartment E_Q contains traced exposed individuals, while Q contains quarantined individuals assumed not to contribute to transmission.

Contact tracing is parameterised through tracing efficiency (η), population compliance (ψ), and tracing delay (τ_{CT}). These quantities determine the probability that exposed contacts are identified and quarantined before generating secondary infections. Analytical expressions for the effective reproduction number were derived for both isolation and contact-tracing scenarios and were subsequently compared with stochastic network simulations.

The deterministic formulation was used to derive analytical thresholds and intervention benchmarks, whereas the corresponding network implementation was used to evaluate how these conclusions changed under explicit contact structure and stochastic transmission.

Disease-induced mortality was added as an outcome layer. A fixed case-fatality ratio (CFR) was applied to individuals leaving the symptomatic infectious compartment, so that a fraction CFR of removed individuals represents deaths (compartment D) and the remainder recoveries. Because deaths and recoveries are both non-transmitting, this partition does not alter the transmission dynamics or the attack rates; it only rescales

the removed pool into fatal and non-fatal outcomes. The baseline CFR was set to 0.32, the case-fatality observed in the Epuyén ANDV outbreak [2]; reported values range from about 21% to 50% across Argentine and Chilean HCPS series [2,7], and this range was used in sensitivity checks.

2.3 Intervention scenarios and simulation design

Four intervention settings are considered: (i) a baseline outbreak without control measures, (ii) symptomatic isolation following diagnosis, (iii) symptomatic isolation combined with contact tracing and quarantine, and (iv) ring quarantine of contacts around diagnosed cases.

These scenarios represent progressively stronger public-health responses and allow the relative contributions of isolation, tracing, and broader quarantine measures to be assessed.

The analysis combined deterministic and stochastic modelling approaches. SEIR and SEIQR models were used to characterise epidemic trajectories, reproduction-number thresholds, and control conditions. Stochastic simulations were then performed on network to account for heterogeneity, clustering, and stochastic extinction following a single imported case.

For the network analysis, outbreak risk was quantified using repeated Monte Carlo simulations. Outcomes included final attack rate, peak prevalence, effective reproduction numbers, and the probability that a single introduction generated a major epidemic. Additional sensitivity and robustness analyses were performed to evaluate uncertainty in epidemiological parameters and alternative network structures.

Together, these analyses allow the three research questions to be examined from both a threshold-based and a risk-based perspective.

3. Results

3.1 Baseline epidemic dynamics

R_0The calibrated SEIR model was first analysed in the absence of intervention to establish a reference epidemic trajectory. Model parameters were chosen to reproduce the pre-intervention transmissibility observed during the Epuyén outbreak (R_0 = 2.12) [2] while retaining the prolonged incubation period characteristic of Andes virus infection.

Under the baseline scenario, the epidemic expanded slowly because of the extended exposed period but ultimately infected a large fraction of the population. The symptomatic infectious compartment reached a peak of approximately 450 individuals around day 186, while the final attack rate was 82.6%. These results indicate that sustained transmission can occur even when epidemic growth is initially gradual.

The long delay before the epidemic peak reflects the unusually long incubation and prodromal phases associated with ANDV. Nevertheless, once transmission became established, the calibrated reproduction number remained sufficiently high to generate a large outbreak. In the deterministic framework, the epidemic therefore does not self-extinguish and provides a useful benchmark against which intervention scenarios can be evaluated.

Although epidemic growth was slow, the model predicts that ANDV can generate a large outbreak if transmission becomes established, highlighting the importance of early intervention.

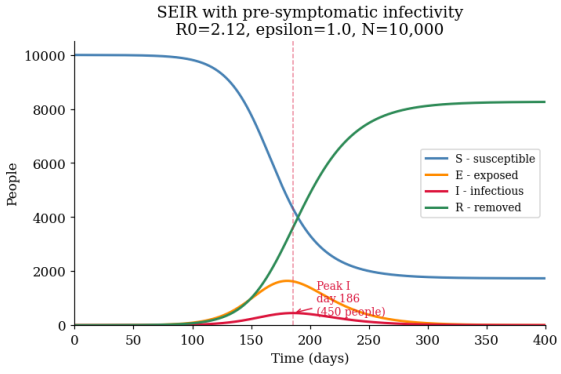


Figure 1. Simulated epidemic trajectory under the no-intervention scenario

3.2 Effect of symptomatic isolation

Symptomatic isolation substantially reduced epidemic burden compared with the no-intervention baseline, particularly when diagnosis occurred rapidly. With a one-day

isolation delay, the epidemic peak decreased from approximately 450 infectious individuals to fewer than 50, while the final attack rate fell from 82.6% to 68.2%. Longer diagnostic delays progressively reduced intervention effectiveness, with attack rates of 76.1% and approximately 80% for five-day and fourteen-day delays, respectively.

These results suggest that rapid diagnosis and isolation remain valuable for reducing epidemic burden and healthcare pressure by suppressing the epidemic peak. However, isolation alone is unlikely to achieve reliable containment when substantial transmission occurs before symptom onset, motivating the evaluation of contact tracing and quarantine strategies.

3.3 Contact tracing and analytical control thresholds

Contact tracing was substantially more effective than symptomatic isolation alone because it targeted infected individuals before symptom onset, interrupting the transmission pathway. Under the baseline assumptions ($\psi = 0.7$ and $\tau_{CT} = 2$ days), increasing tracing efficiency progressively reduced transmission. At the baseline tracing efficiency ($\eta = 0.7$), the controlled reproduction number fell from Reff = 1.89 under isolation alone to $R_{CT} = 1.06$, bringing the system close to the epidemic threshold.

The most important finding was the existence of a clear operational threshold for epidemic control. The critical tracing efficiency required to achieve $R_{CT} < 1$ was approximately $\eta^* = 0.75$. In practical terms, this means that public-health teams would need to identify and quarantine roughly three out of every four relevant contacts of an infected individual before those contacts become infectious themselves. Below this level, transmission remained capable of sustaining epidemic growth; above it, outbreaks could no longer maintain themselves in the mean-field model.

The analysis also showed that tracing speed was nearly as important as tracing coverage. Higher compliance reduced the tracing effort required for control, whereas longer tracing delays increased the threshold and, in some cases, made containment unattainable. From an operational perspective, locating a large proportion of contacts after they have already become infectious provides limited epidemiological benefit. Rapid identification and quarantine are therefore essential for successful outbreak control.

The results therefore suggest that outbreak preparedness should prioritise tracing capacity, public compliance, and rapid response systems rather than relying solely on the isolation of symptomatic cases.

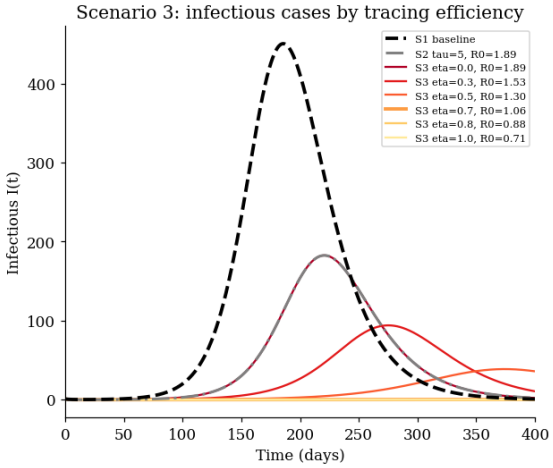


Figure 2. Impact of contact-tracing efficiency on infectious prevalence.

3.4 Global sensitivity analysis

Global sensitivity analysis was used to assess the robustness of the model conclusions to uncertainty in epidemiological and intervention parameters. Across all outcomes examined, the baseline reproduction number (R_0) and tracing efficiency (η) emerged as the dominant drivers of epidemic behaviour.

Higher values of R_0 increased epidemic size and peak prevalence, whereas improved tracing efficiency produced the largest reductions in transmission and outbreak burden. In contrast, uncertainty in compliance (ψ), pre-symptomatic infectiousness (ε), isolation delay (τ_{iso}), and tracing delay (τ_{CT}) had smaller effects within the parameter ranges examined.

The sensitivity analysis reinforces the main conclusions: control was concentrated in scenarios combining moderate transmissibility with high tracing

performance, identifying tracing efficiency as the dominant operational determinant of outbreak control. In practical terms, improving tracing efficiency was more influential than modest improvements in any other controllable intervention parameter.

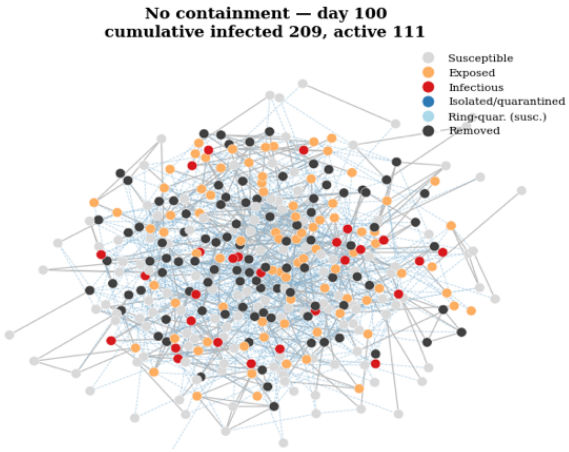
3.5 Synthetic contact network

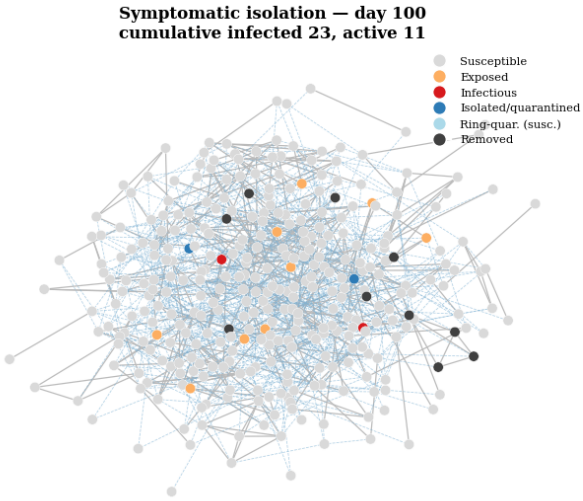
The synthetic network contained 10,000 individuals connected through approximately 30,000 close-contact links, with a mean degree close to six contacts per person. The resulting structure displayed clustering due to household cliques and repeated close-contact interactions. Degree distributions showed moderate heterogeneity, while extra-household links created pathways through which infection could spread between households. These characteristics reproduce the contact patterns most relevant to documented ANDV transmission and provide a realistic framework for evaluating stochastic outbreaks and contact-tracing interventions.

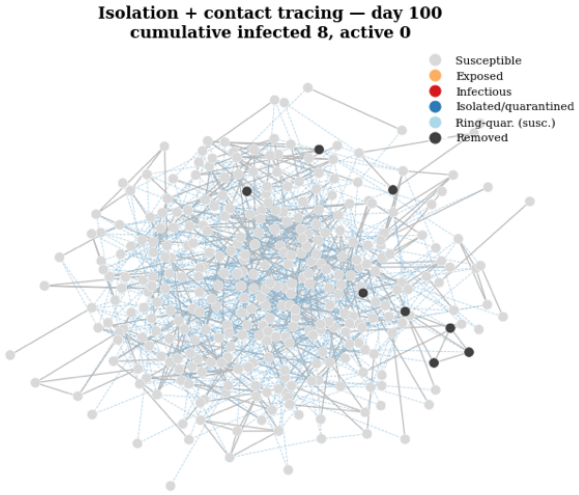
3.6 Network simulation of intervention scenarios

The deterministic analyses identified the conditions required for epidemic control under average population behaviour. In practice, however, outbreaks begin with individual transmission events and are strongly influenced by chance. The network simulations therefore addressed a more practical question: if a single infected traveller introduces ANDV into the community, what is the probability that a major outbreak will actually occur?

Under baseline conditions, a major epidemic developed in 39% of simulations. In other words, most introductions failed to generate sustained transmission, but when outbreaks became established they remained substantial, producing a mean final attack rate of 32.3%. These findings highlight the importance of stochastic extinction: the epidemic is not guaranteed to spread following every introduction, yet the risk of a large outbreak remains sufficiently high to justify intervention. Figure 3 shows the outbreak state on the contact network under the different scenarios.







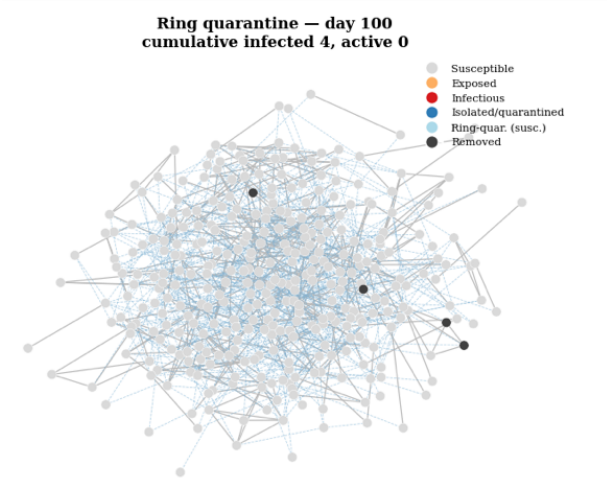


Figure 3. Outbreak state on the close-contact network at day 100 under different intervention scenarios.

Symptomatic isolation substantially reduced both outbreak probability and epidemic burden. The probability of a major outbreak fell from 39% to 22%, while the mean final attack rate decreased from 32.3% to 17.1%. From a public-health perspective, this means that rapid identification and isolation of cases can nearly halve both outbreak frequency and outbreak size, even when complete containment is not achieved.

The addition of contact tracing produced further reductions. Under the baseline tracing assumptions, the probability of a major epidemic decreased to 18% and the mean final attack rate fell to 11.5%. Relative to the no-intervention baseline, this corresponds to a reduction of more than 50% in outbreak risk and approximately two-thirds in epidemic size. Importantly, these simulations confirm the message emerging from the deterministic threshold analysis: interventions that act before symptom onset are substantially more effective than those that rely solely on the isolation of symptomatic individuals.

Ring quarantine produced the strongest overall intervention effect. Extending quarantine measures to all identified close contacts reduced the probability of a major epidemic to 17% and lowered the mean final attack rate to below 10%. However, the improvement relative to standard contact tracing was modest. This suggests that an important part of the achievable benefit is already captured by efficient tracing and quarantine of high-risk contacts.

Applying the baseline case-fatality ratio (CFR) of 0.32 to the simulated outbreaks yields approximately 1,030, 550, 370, and 310 deaths per 10,000 residents under the baseline, symptomatic-isolation, contact-tracing, and ring-quarantine scenarios, respectively. Mortality would be higher in simulations resulting in a major outbreak. Because deaths increase in proportion to the number of cases, the relative effectiveness of the interventions remains unchanged. However, these estimates highlight that even the outbreaks remaining after control measures could still result in a substantial number of deaths. The most important message from the network simulations is that no reactive intervention achieved guaranteed containment once local transmission had begun. Even highly effective tracing and quarantine strategies left a residual probability of a major outbreak because some transmission occurred before cases could be detected and contacts isolated. Nevertheless, each additional layer of intervention produced meaningful reductions in both outbreak probability and epidemic burden.

From a policy perspective, the results suggest that outbreak control should be viewed as a process of risk reduction rather than absolute prevention. Rapid case isolation, efficient contact tracing, and targeted quarantine can substantially reduce the likelihood and size of outbreaks, but they cannot fully compensate for delays introduced by pre-symptomatic transmission. Consequently, preparedness plans should prioritise rapid detection, high tracing coverage, and immediate quarantine of high-risk contacts whenever person-to-person transmission of ANDV is suspected.

3.6 Network simulation of intervention scenarios Comparing deterministic and network framework

The deterministic and network models provided complementary perspectives on Andes virus transmission and control. The deterministic framework described average epidemic behaviour under homogeneous

mixing assumptions and enabled the derivation of analytical control thresholds, including the critical levels of pre-symptomatic infectiousness (ε*) and contact-tracing efficiency (η*) required for epidemic control. These results clarified the mechanisms through which different interventions influence transmission and provided a useful baseline for policy evaluation.

In contrast, the network model explicitly represented household clustering, heterogeneous contact patterns, and stochastic transmission events. This allowed epidemic outcomes to vary between simulation runs and revealed an important feature that cannot be captured by deterministic models: many introductions failed to establish sustained transmission even when the average reproduction number exceeded one. As a result, epidemic risk was more appropriately characterised by both the probability of a major outbreak and the distribution of outbreak sizes rather than by a single average epidemic trajectory.

Quantitatively, the network simulations produced lower average attack rates than the deterministic model because transmission chains frequently died out during the early stages of an outbreak. Nevertheless, once a major epidemic became established, the qualitative behaviour remained consistent with the deterministic predictions. Symptomatic isolation reduced epidemic burden but rarely achieved reliable containment on its own, whereas contact tracing and quarantine generated the largest reductions in transmission and outbreak probability.

Overall, the agreement between the two modelling frameworks strengthens confidence in the main conclusions. The deterministic model provides analytical insight into intervention thresholds and epidemic dynamics, while the network model demonstrates that these conclusions remain robust when realistic contact structure and stochastic effects are incorporated.

3.7 Robustness to network structure

Additional analyses were performed to assess whether the results depended on the specific contact network used in the baseline simulations. Alternative network configurations varied both the average number of extra-household contacts and the degree of contact heterogeneity.

Although these modifications altered network properties and transmission potential, the qualitative conclusions remained unchanged. Across all network variants, symptomatic isolation reduced epidemic burden, contact tracing provided additional reductions in outbreak risk, and ring quarantine remained the most effective intervention.

These findings indicate that the principal conclusions are robust to plausible variations in contact-network structure and are not driven by a particular network specification.

4. Discussion

4.1 Limitations

First, the epidemiological parameters were transferred from documented Argentine and Chilean ANDV outbreaks, principally the 2018–2019 Epuyén cluster [2], rather than estimated from Italian data, and the relative pre-symptomatic infectiousness was fixed at $\varepsilon = 1$ in the absence of a quantitative estimate. These choices are conservative, and the sensitivity and robustness analyses indicate that the qualitative conclusions do not hinge on their exact values; nevertheless, the absolute figures should be read as illustrative rather than predictive.

Second, the contact network is synthetic rather than empirically observed. Although it reproduces key features of close-contact transmission, including household clustering and age-assortative mixing, it cannot capture all aspects of real social networks. Nevertheless, robustness analyses indicated that the principal intervention ranking remained unchanged across alternative network structures.

Third, mortality is represented only through a fixed case-fatality ratio applied to cases; healthcare burden, disease severity, and age- or comorbidity-dependent fatality were not

explicitly modelled, although ANDV hantavirus cardiopulmonary syndrome carries a high and heterogeneous case-fatality rate.

Finally, the study assumed no adaptive behavioural response. In practice, public awareness, voluntary contact reduction, and changes in healthcare-seeking behaviour could influence both outbreak dynamics and intervention effectiveness.

5. Conclusions

This study examined the potential spread of Andes virus following a hypothetical introduction into an Italian coastal community and evaluated the effectiveness of progressively stronger intervention strategies.

The first research question concerned epidemic behaviour in the absence of intervention. Both deterministic and network simulations indicated that ANDV has the potential to generate outbreaks because of its long incubation period and person-to-person transmission. Although many introductions failed to establish transmission in the stochastic network simulations, large outbreaks remained possible once transmission became established.

The second research question examined the effectiveness of symptomatic isolation. Rapid isolation reduced epidemic burden by lowering both peak prevalence and final attack rate. However, isolation alone was unable to achieve reliable containment because transmission occurring before symptom onset remained sufficient to sustain epidemic growth. The threshold $\varepsilon^* \approx 0.60$ demonstrated that this conclusion is robust across a broad range of assumptions regarding pre-symptomatic infectiousness.

The third research question focused on contact tracing and quarantine. Contact tracing emerged as the most important reactive intervention because it directly targeted transmission before symptom onset. Under the baseline assumptions, epidemic control required tracing efficiency to exceed $\eta^* \approx 0.75$, corresponding to the successful identification and quarantine of roughly three out of every four relevant contacts. Network simulations confirmed that tracing substantially reduced both outbreak probability and epidemic size, although no reactive intervention eliminated outbreak risk entirely.

Because ANDV is highly lethal, even relatively small outbreaks can result in substantial mortality. Consequently, reducing outbreak probability is important not only for limiting transmission but also for preventing an important number of deaths.

Taken together, the results suggest three practical implications. First, symptomatic isolation should be viewed primarily as a mitigation measure rather than a stand-alone containment strategy. Second, tracing efficiency, compliance, and tracing speed are the key operational determinants of outbreak control. Third, because transmission is concentrated within households and other close-contact settings, interventions targeting local contact clusters are likely to provide the greatest epidemiological benefit.

Overall, the study suggests that the control of person-to-person ANDV transmission depends less on the isolation of symptomatic cases than on the ability to rapidly identify and quarantine exposed contacts before they become infectious. Across deterministic analyses, stochastic network simulations, and robustness tests, the same conclusion emerged: interventions that act before symptom onset are consistently more effective than those that act afterwards.

Although the results are not intended to predict a real outbreak, they show how transmission before symptoms, social contacts, and the effectiveness of control measures can affect outbreak outcomes.

References

[1] Martinez, V.P., Bellomo, C., San Juan, J., Pinna, D., Forlenza, R., Elder, M., Padula, P.J. (2005). Person-to-person transmission of Andes virus. Emerging Infectious Diseases 11(12), 1848–1853.

[2] Martinez, V.P., Di Paola, N., Alonso, D.O., Perez-Sautu, U., Bellomo, C.M., Iglesias, A.A., et al. (2020). Super-spreaders and person-to-person transmission of Andes virus in Argentina. New England Journal of Medicine 383, 2230–2241.

[3] Martinez-Valdebenito, C., Calvo, M., Vial, C., Mansilla, R., Marco, C., Palma, R.E., Vial, P.A., Valdivieso, F., Mertz, G., Ferres, M. (2014). Person-to-person household and nosocomial transmission of Andes hantavirus, southern Chile, 2011. Emerging Infectious Diseases 20(10), 1629–1636.

[4] Rizi, A.K., Keating, L.A., Gleeson, J.P., O'Sullivan, D.J.P., Kivela, M. (2024). Effectiveness of contact tracing on networks with cliques. Physical Review E 109, 024303.

[5] World Health Organization (2026). Hantavirus cluster linked to cruise ship travel, multi-country. Disease Outbreak News, 4, 8 and 13 May 2026.

[6] Vial, P.A., Valdivieso, F., Mertz, G., Castillo, C., Belmar, E., Delgado, I., Tapia, M., Ferres, M. (2006). Incubation period of hantavirus cardiopulmonary syndrome. Emerging Infectious Diseases 12(8), 1271–1273.

[7] Ferres, M., Vial, P., Marco, C., Yanez, L., Godoy, P., Castillo, C., Hjelle, B., Delgado, I., Lee, S.-J., Mertz, G.J. (2007). Prospective evaluation of household contacts of persons with hantavirus cardiopulmonary syndrome in Chile. Journal of Infectious Diseases 195(11), 1563–1571.

[8] Galeno, H., Mora, J., Villagra, E., Fernandez, J., Hernandez, J., Mertz, G.J., Ramirez, E. (2002). First human isolate of hantavirus (Andes virus) in the Americas. Emerging Infectious Diseases 8(7), 657–661.

[9] Melegaro, A., Jit, M., Gay, N., Zagheni, E., Edmunds, W.J. (2011). What types of contacts are important for the spread of infections? Epidemics 3(3–4), 143–151.

[10] Grimm, V., Mengel, F., Schmidt, M. (2021). Extensions of the SEIR model for the analysis of tailored social distancing and tracing approaches to cope with COVID-19. Scientific Reports 11, 4214.

[11] Reyna-Lara, A., Soriano-Panos, D., Gomez, S., Granell, C., Matamalas, J.T., Steinegger, B., Arenas, A., Gomez-Gardenes, J. (2021). Virus spread versus contact tracing: two competing contagion processes. Physical Review Research 3, 013163.

[12] Istituto Nazionale di Statistica (ISTAT) (2022). Censimento permanente della popolazione e delle abitazioni: popolazione residente e dinamica demografica, anno 2021. ISTAT, Roma. http://dati-censimentipermanenti.istat.it/

[13] Prem, K., van Zandvoort, K., Klepac, P., Eggo, R.M., Davies, N.G., Cook, A.R., Jit, M. (2021). Projecting contact matrices in 177 geographical regions: an update and comparison with empirical data for the COVID-19 era. PLoS Computational Biology 17(7), e1009098.

[14] Fumanelli, L., Ajelli, M., Manfredi, P., Vespignani, A., Merler, S. (2012). Inferring the structure of social contacts from demographic data in the analysis of infectious diseases spread. PLoS Computational Biology 8(9), e1002673.

[15] Iozzi, F., Trusiano, F., Chinazzi, M., Billari, F.C., Zagheni, E., Merler, S., Ajelli, M., Del Fava, E., Manfredi, P. (2010). Little Italy: an agent-based approach to the estimation of contact patterns. PLoS Computational Biology 6(12), e1001021.